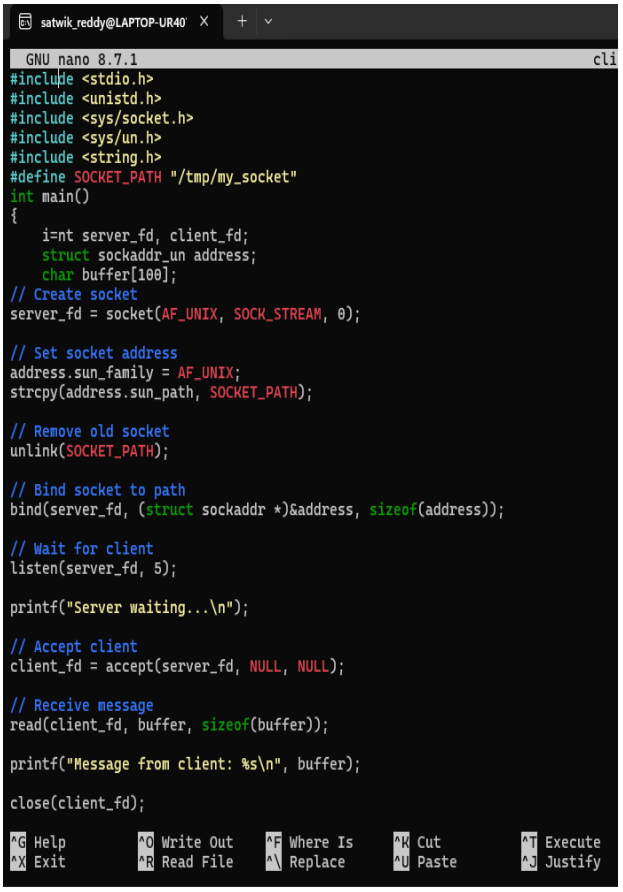
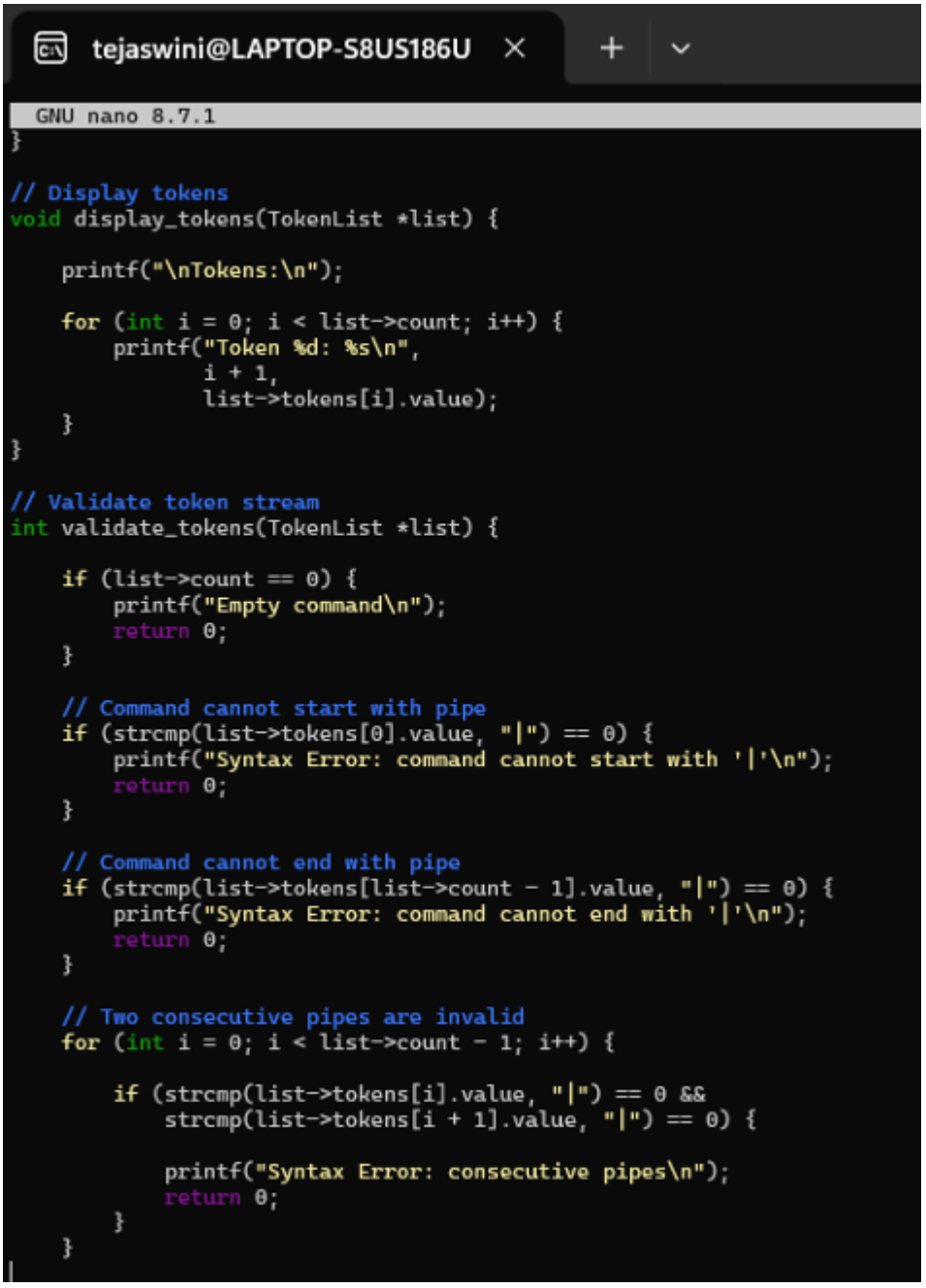
PRACTICAL

OSSP - SEC2




tejaswini@LAPTOP-S8US186U
GNU nano 8.7.1
}
// Display tokens
void display_tokens(TokenList *list) {
printf("\nTokens:\n");
for (int i = 0; i < list->count; i++) {
printf("Token %d: %s\n",
i + 1,
list->tokens[i].value);
}
}
// Validate token stream
int validate_tokens(TokenList *list) {
if (list->count == 0) {
printf("Empty command\n");
return 0;
}
// Command cannot start with pipe
if (strcmp(list->tokens[0].value, "|") == 0) {
printf("Syntax Error: command cannot start with '|'\n");
return 0;
}
// Command cannot end with pipe
if (strcmp(list->tokens[list->count - 1].value, "|") == 0) {
printf("Syntax Error: command cannot end with '|'\n");
return 0;
}
// Two consecutive pipes are invalid
for (int i = 0; i < list->count - 1; i++) {
if (strcmp(list->tokens[i].value, "|") == 0 &&
strcmp(list->tokens[i + 1].value, "|") == 0) {
printf("Syntax Error: consecutive pipes\n");
return 0;
}
}

```
  GNU nano 8.7.1

// Simple parse tree representation
void create_parse_tree(TokenList *list) {

    printf("\nParse Tree:\n");

    printf("COMMAND\n");

    for (int i = 0; i < list->count; i++) {

        if (strcmp(list->tokens[i].value, "|") == 0) {
            printf("  PIPE\n");
        }
        else if (strcmp(list->tokens[i].value, ">") == 0) {
            printf("  OUTPUT_REDIRECTION\n");
        }
        else if (strcmp(list->tokens[i].value, "<") == 0) {
            printf("  INPUT_REDIRECTION\n");
        }
        else {
            printf("  ARGUMENT: %s\n",
                   list->tokens[i].value);
        }
    }
}

// Free allocated memory
void free_tokens(TokenList *list) {

    for (int i = 0; i < list->count; i++) {
        free(list->tokens[i].value);
    }

    list->count = 0;
}

int main() {

    char input[MAX_INPUT];

    TokenList list;
```

```c
    // Remove newline
    input[strcspn(input, "\n")] = '\0';

    // Handle empty command
    if (strlen(input) == 0) {
        printf("Empty command\n");
        return 0;
    }

    tokenize(input, &list);

    display_tokens(&list);

    if (validate_tokens(&list)) {

        printf("\nSyntax is valid.\n");

        create_parse_tree(&list);

        printf("\nExecution structure created successfully.\n");
    }
    else {
        printf("\nInvalid command.\n");
    }

    free_tokens(&list);

    return 0;
}
```

Output

```
satwik@LAPTOP-S8US186U:~$ gcc skill_4.c -o skill_4
satwik@LAPTOP-S8US186U:~$ ./skill_4
Enter a command: ls -l

Tokens:
Token 1: ls
Token 2: -l
Token 2: -l

Syntax is valid.

Parse Tree:
   COMMAND
      ARGUMENT: ls
      ARGUMENT: -l
      ARGUMENT:

Execution structure created successfully.
```

```
satwik@LAPTOP-S8US186U:~$ gcc skill_4.c -o skill_4
satwik@LAPTOP-S8US186U:~$ ./skill_4
Enter a command: ls || grep txt

Tokens:
Token 1: ls
Token 2: ||
Token 3: grep
Token 4: txt

Syntax is valid.

Parse Tree:
   COMMAND
      ARGUMENT: ls
      PIPE
      ARGUMENT: grep
      ARGUMENT: txt

Execution structure created successfully.
```

```
satwik@LAPTOP-S8US186U:~$ gcc skill_4.c -o skill_4
satwik@LAPTOP-S8US186U:~$ ./skill_4
Enter a command: ls || grep

Tokens:
Token 1: ls
Token 2: ||
Token 3: grep

Syntax is valid.

Parse Tree:
   COMMAND
      ARGUMENT: ls
      PIPE
      ARGUMENT: grep

Execution structure created successfully.
```